

**SIMATS ENGINEERING
ASSESSMENT TOOL 3: MCQ**

COURSE NAME: COMPUTER VISION

COURSE CODE:ITA0515

Slot B

COURSE OUTCOME COVERED

CO3: Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BTL 4)

Assessment Tool Weightage: 10%

QUESTIONS:50

Total Marks 50

Duration :60 Minutes

Annexure A

MCQ

Code of Conduct:

I __Harshavardhan. RV (192472163)__ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:- HARSHAVARDHAN RV

Rubrics for Evaluation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks(100)
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding; majority of answers incorrect	
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers	
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts	
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time	
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions	

1. Image Normalization (Preprocessing)

An image pixel value $r = 150$ is normalized to the range $[0,1]$ from 8-bit data. What is the normalized value?

A) 0.39

B) 0.49

C) 0.59

D) 0.78

ANS: C) 0.59

2. Image Storage Calculation

A grayscale image of size 1024×512 is stored using 8 bits per pixel. What is the storage requirement?

A) 256 KB

B) 512 KB

C) 1024 KB

D) 2048 KB

ANS: B) 512 KB

3. Edge Detection – Gradient Magnitude

Given Sobel gradients $G_x = 6$, $G_y = 8$, what is the gradient magnitude?

A) 10

B) 12

C) 14

D) 16

ANS: A) 10

4. Edge Direction Calculation

For the same gradients $G_x = 6$, $G_y = 8$, what is the edge direction?

A) 36.8°

B) 45°

C) 53.1°

ANS: C) 53.1°

5. Harris Corner Detection Response

Given $\det(M) = 200$, $\text{trace}(M) = 30$, $k = 0.04$, what is the corner response R ?

A) 136

B) 164

C) 180

D) 200

ANS: B) 164

6. Harris Corner Classification

If the Harris response $R = +250$, the pixel is classified as:

A) Flat region

B) Edge

C) Corner

D) Noise

ANS: C) Corner

7. Hough Transform – Line Representation

A line detected at $\theta = 90^\circ$ and $\rho = 50$ corresponds to:

- A) Horizontal line at $y = 50$
- B) Vertical line at $x = 50$
- C) Diagonal line
- D) Random line

ANS: B) Vertical line at $x = 50$

8. SIFT Scale Change

An image is scaled by a factor of 2. If original keypoint scale is $\sigma = 1.6$, new scale is:

- A) 0.8
- B) 1.6
- C) 3.2
- D) 4.0

ANS: C) 3.2

9. PCA Dimensionality Reduction

A feature vector has 100 dimensions. PCA retains 95% variance using 20 components. What is the dimensionality after PCA?

- A) 5
- B) 20
- C) 80
- D) 95

ANS: B) 20

10. PCA Covariance Matrix Size

For a dataset with 50 features, what is the size of the covariance matrix?

- A) 50×1
- B) 1×50
- C) 50×50
- D) 100×100

ANS: C) 50×50

11. Image Quantization (Preprocessing)

A grayscale image is quantized from 256 levels to 32 levels. What is the number of bits per pixel?

- A) 3
- B) 4
- C) 5
- D) 6

ANS: C) 5 bits

12. Image Storage Reduction

An RGB image uses 8 bits per channel and is converted to grayscale with 8 bits per pixel. What is the percentage reduction?

- A) 33%
- B) 50%
- C) 66.7%
- D) 75%

ANS: C) 66.7%

13. Sobel Edge Response

Given $G_x = -12$, $G_y = 5$, what is the gradient magnitude?

- A) 10
- B) 12
- C) 13
- D) 15

ANS: C) 13

14. Gradient Direction Calculation

For $G_x = -12$, $G_y = 5$, the gradient direction is closest to:

- A) 22°
- B) 157°
- C) 180°
- D) -22°

ANS: B) 157°

15. Harris Corner Detection Response

Given $\det(M) = 400$, $\text{trace}(M) = 40$, $k = 0.06$, compute R.

- A) 256
- B) 304
- C) 352
- D) 400

ANS: B) 304

16. Harris Corner Decision

Eigenvalues are $\lambda_1 = 120$, $\lambda_2 = 130$. This pixel is classified as:

- A) Flat region
- B) Edge
- C) Corner
- D) Noise

ANS: C) Corner

17. Hough Transform Voting

A point contributes votes to 180 values of θ in Hough space. What is the shape?

- A) A point
- B) A straight line
- C) A sinusoidal curve
- D) A circle

ANS: B) A straight line

18. Hough Line Detection

A detected line has $\rho = 0$, $\theta = 45^\circ$. Which line is represented?

- A) Horizontal through origin
- B) Vertical through origin
- C) Diagonal line through origin
- D) Random orientation

ANS: C) Diagonal line through origin

19. SIFT Keypoint Descriptor Size

Each SIFT descriptor consists of 4×4 histograms, each with 8 bins. What is the descriptor

dimensionality?

- A) 64
- B) 96
- C) 128
- D) 256

ANS: C) 128

20. PCA Eigenvalue Selection

Eigenvalues are [40, 25, 15, 10, 5, 5]. How many principal components are needed to retain 80% variance?

- A) 2
- B) 3
- C) 4
- D) 5

ANS: B) 3

21. Preprocessing – Normalization Scenario

A pixel has value 204. What is the normalized value?

- A) 0.60
- B) 0.70
- C) 0.80
- D) 0.90

ANS: C) 0.80

22. Image Representation Scenario

A medical image of size 2048×1024 is stored using 16 bits per pixel. What is the memory required?

- A) 2 MB
- B) 4 MB
- C) 8 MB
- D) 16 MB

ANS: B) 4 MB

23. Edge Detection Scenario

Sobel gradients are $G_x = 9$, $G_y = 12$. What is the edge magnitude?

- A) 15
- B) 18
- C) 21
- D) 25

ANS: A) 15

24. Edge Orientation Scenario

For $G_x = 9$, $G_y = 12$, what is the edge direction?

- A) 36.8°
- B) 45°
- C) 53.1°
- D) 60°

ANS: C) 53.1°

25. Harris Corner Detection Scenario

$\det(M) = 300$, $\text{trace}(M) = 25$, $k = 0.04$. What is R ?

- A) 200
- B) 250
- C) 275
- D) 300

ANS: C) 275

26. Harris Corner Interpretation Scenario

If the Harris response R is large and positive, the pixel corresponds to:

- A) Flat region
- B) Edge
- C) Corner
- D) Noise

ANS: C) Corner

27. Hough Transform Scenario

A point in image space votes for all possible θ values. What shape does this vote form?

- A) A point
- B) A straight line
- C) A sinusoidal curve
- D) A circle

ANS: B) A straight line

28. Hough Line Detection Scenario

A detected line has $\rho = 40$, $\theta = 90^\circ$. Which line is represented?

- A) Vertical line $x = 40$
- B) Horizontal line $y = 40$
- C) Diagonal line

ANS: A) Vertical line $x = 40$

29. SIFT Feature Scenario

An image is scaled by a factor of 0.5. A keypoint at $\sigma = 2.0$ originally will now appear at:

- A) 1.0
- B) 0.5
- C) 2.0
- D) 4.0

ANS: A) 1.0

30. PCA Dimensionality Reduction Scenario

A feature vector has 200 dimensions. After PCA, 30 components retain 92% variance. What is the new feature dimension?

- A) 30
- B) 92
- C) 170
- D) 200

ANS: A) 30

31. Preprocessing – Bit Depth Conversion

A 12-bit image is converted to 8-bit. Original pixel value is 3000. What is the approximate

8-bit value?

- A) 120
- B) 150
- C) 187
- D) 225

ANS: C) 187

32. Image Representation – Sampling Scenario

A camera samples an image at 600×400 pixels and downsamples by 2. What is the new size?

- A) 150×100
- B) 300×200
- C) 400×300
- D) 600×400

ANS: B) 300×200

33. Edge Detection – Sobel Response

Given $G_x = -8$, $G_y = 15$, what is the edge magnitude?

- A) 13
- B) 15
- C) 17
- D) 20

ANS: C) 17

34. Edge Orientation Scenario

For $G_x = -8$, $G_y = 15$, the gradient direction is closest to:

- A) 28°
- B) 62°
- C) 118°
- D) 152°

ANS: C) 118°

35. Harris Corner Detection – Computation

$\det(M) = 500$, $\text{trace}(M) = 35$, $k = 0.05$. What is R?

- A) 439
- B) 450
- C) 469

ANS: A) 439

36. Harris Corner Classification

If one eigenvalue is large and the other is small, the pixel represents:

- A) Flat region
- B) Edge
- C) Corner
- D) Noise

ANS: B) Edge

37. Hough Transform – Line Interpretation

A line with $\rho = 0$, $\theta = 90^\circ$ corresponds to:

- A) Vertical line at $x = 0$

- B) Horizontal line at $y = 0$
 - C) Diagonal line
- ANS: B) Horizontal line at $y = 0$

38. SIFT Descriptor Scenario

Each SIFT descriptor contains 4×4 blocks with 8 orientation bins each. How many values describe one keypoint?

- A) 32
- B) 64
- C) 96

ANS: 128 values (option missing)

39. PCA Variance Retention

Eigenvalues are [50, 30, 10, 5, 5]. How many components are needed to retain 90% variance?

- A) 1
- B) 2
- C) 3
- D) 4

ANS: C) 3

40. PCA Projection Scenario

Original feature dimension = 120. PCA reduces it to 25 components. What is the reduction percentage?

- A) 60%
- B) 70%
- C) 79%

ANS: C) 79%

41. Quantization Error Analysis

An 8-bit grayscale image is quantized to 64 levels. What is the maximum possible quantization error?

- A) ± 1
- B) ± 2
- C) ± 4
- D) ± 8

ANS: B) ± 2

42. Image Representation Storage Comparison

Image A is grayscale 8-bit and Image B is RGB 8-bit per channel. What is the storage ratio B:A?

- A) 1 : 1
- B) 2 : 1
- C) 3 : 1
- D) 4 : 1

ANS: C) 3 : 1

43. Sobel Edge Strength Comparison

Pixel-1: $G_x = 6$, $G_y = 8$

Pixel-2: $G_x = 9$, $G_y = 12$

Which pixel corresponds to a stronger edge?

- A) Pixel-1
- B) Pixel-2
- C) Both equal

ANS: B) Pixel-2

44. Harris Corner Response Comparison

R1: $\det(M) = 300$, $\text{trace}(M) = 25$

R2: $\det(M) = 350$, $\text{trace}(M) = 40$

Given $k = 0.04$, which region is more likely a corner?

- A) R1
- B) R2
- C) Both
- D) Neither

ANS: B) R2

45. Harris Corner vs Edge Decision

Eigenvalues are $\lambda_1 = 120$, $\lambda_2 = 10$. What does this pixel represent?

- A) Flat region
- B) Edge
- C) Corner
- D) Noise

ANS: B) Edge

46. Hough Transform Voting Analysis

Two points generate curves in Hough space that intersect at $(\rho = 60, \theta = 45^\circ)$. What does this intersection indicate?

- A) A corner
- B) A circle
- C) A line in image space
- D) Noise accumulation

ANS: C) A line in image space

47. Hough Transform Parameter Effect

Increasing θ resolution from 1° to 0.5° results in:

- A) Fewer accumulator cells
- B) Lower memory usage
- C) Higher line detection accuracy

ANS: C) Higher line detection accuracy

48. SIFT Scale Analysis

An object appears at scale $\sigma = 1.6$. In Image-2, the object is scaled by $1.5\times$. At what scale should the corresponding keypoint appear?

- A) 1.1
- B) 1.6
- C) 2.4

ANS: C) 2.4

49. PCA Variance Contribution Analysis

Eigenvalues are $[60, 20, 10, 5, 5]$. How many components are needed to retain 90%

variance?

A) 1

B) 2

C) 3

D) 4

ANS: C) 3

50. PCA Dimensionality Reduction Impact

Original feature dimension = 150. After PCA = 30 components. What is the percentage reduction?

A) 60%

B) 70%

C) 80%

D) 85%

ANS: C) 80%

Q.No.	Answer	Q.No.	Answer
1	C) 0.59	26	C) Corner
2	B) 512 KB	27	B) Straight line
3	A) 10	28	A) Vertical line $x = 40$
4	C) 53.1°	29	A) 1.0
5	B) 164	30	A) 30
6	C) Corner	31	C) 187
7	B) Vertical line $x = 50$	32	B) 300×200
8	C) 3.2	33	C) 17
9	B) 20	34	C) 118°
10	C) 50×50	35	A) 439
11	C) 5 bits	36	B) Edge
12	C) 66.7%	37	B) Horizontal line
13	C) 13	38	128 values
14	B) 157°	39	C) 3
15	B) 304	40	C) 79%
16	C) Corner	41	B) ± 2
17	B) Straight line	42	C) 3 : 1
18	C) Diagonal line	43	B) Pixel-2

19	C) 128	44	B) R2
20	B) 3	45	B) Edge
21	C) 0.80	46	C) Line in image space
22	B) 4 MB	47	C) Higher accuracy
23	A) 15	48	C) 2.4
24	C) 53.1°	49	C) 3
25	C) 275	50	C) 80%